

Research paper

SiCLIP: An explainable multimodal framework for silicosis diagnosis

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ABSTRACT

Silicosis is a serious occupational lung disease caused by exposure to crystalline silica dust and remains difficult to detect early in at-risk worker populations. In this paper, we introduce the Silicosis Diagnosis Dataset (SDD), which comprises chest X-ray images and structured patient-profile information, including harmful habits and clinical symptoms. To exploit this multimodal dataset, we propose SiCLIP, a multimodal retrieval framework based on CLIP-ViT for silicosis screening and binary classification on SDD. SiCLIP learns a shared embedding space for chest X-ray images and patient profiles and performs retrieval-based aggregation for prediction. On the internally evaluated SDD benchmark, SiCLIP achieves higher accuracy and F1-score than several strong image-only deep learning baselines and the compared multimodal VLM baseline. In addition, SiCLIP provides case-based interpretability by grounding predictions in retrieved similar cases, complemented by supportive saliency visualizations. These results suggest that multimodal retrieval is a promising approach for silicosis screening support in occupationally exposed populations, while external validation remains necessary before broader clinical deployment.

1. Introduction

Silicosis is a chronic occupational pneumoconiosis that develops after repeated or prolonged inhalation of respirable crystalline silica dust, which may originate from materials such as quartz-containing rock, sand, or industrial minerals [1]. Workers in mining, quarrying, stone processing, construction, foundries, glass production, ceramics, and related occupations are therefore among the populations most frequently exposed to this hazard. In coal-mining environments, silica exposure may also coexist with other mineral dusts, increasing the risk of mixed forms of occupational lung disease [2,3]. After deposition in the distal airways and lung parenchyma, silica particles can induce persistent inflammation and progressive fibrotic changes. As the disease advances, patients may experience dyspnea, chronic cough, chest discomfort, fatigue, reduced physical tolerance, and, in severe cases, respiratory impairment that can be life-threatening [4].

The public-health impact of silicosis is especially concerning in low- and middle-income settings, where many high-risk workers have limited access to specialized occupational-health services and advanced imaging facilities. Miners, quarry workers, construction laborers, and

other manual workers may experience substantial cumulative exposure during daily work, yet early disease can remain clinically silent or present with non-specific symptoms. These factors make timely screening and risk stratification difficult in routine practice.

Because symptoms alone are not sufficient to confirm silicosis, clinical assessment usually combines medical examination, occupational exposure history, and radiological evaluation. Physicians typically review the duration and intensity of silica exposure, respiratory complaints, and functional limitations before deciding whether additional tests are required. Chest radiography and computed tomography are commonly used to assess lung abnormalities associated with silica-related disease [5–7]. Pulmonary function tests can provide complementary information about respiratory impairment [8]. In selected cases, bronchoscopy [9], sputum examination [10], or lung biopsy [11] may be considered to support differential diagnosis or further characterize suspicious findings.

Among the available diagnostic tools, chest radiography remains the most practical modality for initial screening and longitudinal monitoring of silicosis in occupationally exposed populations [12]. Our focus is on CXR for several compelling reasons: First, its widespread availability

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and relatively low cost make it highly suitable for large-scale occupational health programs, unlike more expensive modalities like CT scans, which are typically reserved for further investigation [13]. Second, CXR is non-invasive and often mandated in regular health check-ups for workers in high-risk industries (e.g., mining, construction), ensuring a consistent data stream for monitoring [14]. Third, and critically important for diagnosis, CXR directly visualizes the characteristic structural lung damage caused by silica dust, such as the presence and profusion of nodules and opacities [15]. While indispensable as a first-line tool, relying solely on CXR interpretation presents challenges, including inter-observer variability and the difficulty in detecting subtle early-stage disease [16]. Furthermore, radiological findings alone are often insufficient for a conclusive diagnosis. Therefore, integrating CXR findings with crucial contextual information, particularly the patient's occupational exposure history and clinical profile (profile data), is essential for enhancing diagnostic accuracy and differentiating silicosis from other interstitial lung diseases. Accordingly, our framework is intended not as a standalone diagnostic system, but as a multimodal screening and triage support tool for occupationally exposed populations, helping prioritize suspected silicosis cases for further expert review and follow-up assessment. Importantly, CXR is also the only scalable imaging modality in resource-constrained regions, making it the de facto first-line tool for screening large at-risk populations such as miners and manual laborers. Our method is intentionally designed to align with these real-world screening constraints, enabling multimodal assessment even in settings where CT imaging or specialist interpretation is limited. Presently, there is a lack of research exploring the use of deep learning methodologies to synergistically analyze patient profile information alongside CXR data for the diagnosis of silicosis.

Fortunately, the effective integration of information from chest X-ray images and patient profile data can be achieved through a multimodal retrieval framework. In this study, we develop a novel multimodal retrieval architecture, specifically designed to jointly process chest radiographs and structured patient records, thereby enhancing the accuracy of silicosis identification and diagnosis. The main contributions of this study are summarized as follows:

- **Dataset curation:** We develop a new silicosis diagnosis dataset, named SDD, which includes chest X-ray images together with structured patient profiles describing symptoms, harmful habits, and occupational exposure history. This dataset provides a dedicated benchmark for developing and evaluating multimodal silicosis screening and classification methods.
- **Multimodal retrieval-based classification:** We introduce SiCLIP, a multimodal retrieve-and-vote framework based on CLIP-ViT that jointly leverages chest radiographs and patient profiles. By aligning both modalities in a shared embedding space and performing retrieval-based aggregation, SiCLIP improves silicosis classification performance on SDD relative to the evaluated baselines.
- **Case-based interpretability support:** The retrieve-and-vote design provides case-based transparency by linking each prediction to retrieved prior cases with similar radiographic and profile characteristics. This makes the prediction process easier to inspect and audit within the studied setting.
- **Qualitative visualization support:** We further provide profile-conditioned saliency visualizations to qualitatively illustrate regions emphasized by the model under different patient-profile inputs. These visualizations are intended as supportive interpretation aids rather than as direct causal explanations or evidence of deployment readiness.

2. Related work

2.1. Radiologist-guided methods

Chest radiography has long served as one of the most widely used imaging tools for evaluating pulmonary abnormalities [17]. In routine

clinical practice, CXR images are interpreted by radiologists or trained physicians who examine lung fields, anatomical structures, and abnormal opacity patterns to support diagnostic decision-making. Although this expert-driven workflow remains essential, its performance can be influenced by reader experience, image quality, disease subtlety, and the overlap of radiographic signs across different thoracic conditions. As a result, disagreement between readers may occur, particularly in cases with mild or ambiguous findings [18].

To address these challenges, computational methods have been developed to assist CXR interpretation and improve the consistency of image-based assessment. Computer-aided diagnosis (CAD) systems [19] aim to provide additional support by automatically detecting suspicious findings, extracting quantitative image features, or highlighting regions that may require closer inspection. Such systems are not intended to replace radiologists, but rather to reduce workload and provide complementary evidence during the diagnostic process.

In addition to image-focused CAD approaches, clinical decision support systems have been investigated as a way to combine radiographic observations with non-imaging information, such as patient history, symptoms, and other clinical variables [20]. This direction is particularly relevant for lung diseases whose diagnosis depends not only on visual abnormalities in CXR images but also on contextual information about exposure, risk factors, and clinical presentation.

Another line of work has focused on structured and standardized reporting for chest radiographs [21]. By encouraging consistent terminology and reporting formats, these systems can improve communication between radiologists and clinicians, reduce ambiguity in clinical interpretation, and support more reproducible decision-making across healthcare settings.

2.2. Traditional machine learning methods

The application of conventional machine-learning techniques to CXR-based lung-disease detection has been motivated largely by the limited availability of well-annotated medical images [22,23]. In these approaches, the diagnostic pipeline typically depends on predefined image descriptors extracted from segmented lung regions or selected regions of interest. Chandra et al. [24], for example, analyzed lung-region features and evaluated several classical classifiers, including multilayer perceptron, random forest, sequential minimal optimization, regression-based classification, and logistic regression. Kuo et al. [25] used a set of handcrafted features together with a decision-tree classifier for pneumonia identification in patients with schizophrenia. Yue et al. [26] investigated feature representations derived from chest CT images for lung-disease analysis.

Although these studies demonstrate the usefulness of machine-learning models in medical image classification, their reliance on manually designed features limits their scalability. Such pipelines are often sensitive to preprocessing quality, feature selection, and dataset characteristics. Moreover, handcrafted descriptors may fail to capture subtle or spatially complex radiographic patterns in high-dimensional CXR data. These limitations reduce their robustness when applied to heterogeneous clinical datasets and motivate the use of representation-learning methods that can learn task-specific features directly from data.

2.3. Deep learning based methods

Deep learning techniques—particularly convolutional neural networks (CNNs) have fundamentally transformed medical image analysis by enabling end-to-end feature learning, in contrast to traditional methods that depend on handcrafted descriptors [27–30]. Early efforts by Sharma et al. [31] and Stephen et al. [32] demonstrated the utility of relatively simple CNN architectures for classifying pneumonic chest X-rays, employing data augmentation to mitigate the challenges of

limited training data. Building on this foundation, Rajpurkar et al. introduced CheXNet, a deep CNN model that both predicts multiple lung pathologies and produces localization heat maps to highlight regions of interest [33]. Subsequent work has extended these architectures to thoracic anomaly detection, exploring more powerful backbones for tasks such as pneumothorax identification [34].

In response to the COVID-19 pandemic, Ozturk et al. proposed Darknet for automated detection of SARS-CoV-2 from raw chest X-ray images [35], while AlMamlook et al. designed an ensemble framework that integrates multiple machine-learning classifiers with established CNN models to improve diagnostic consistency [36]. To address variability across imaging sources, domain-adaptation strategies have been introduced: Janizek et al. developed an adversarial optimization approach for cross-domain lung disease classification [37], and Zhang et al. presented a confidence-aware module framing disease detection as a one-class anomaly problem [38]. Complementary to these, Tuncer et al. combined fuzzy-tree transformations with multi-kernel local binary patterns to distinguish COVID-19 from other pneumonic patterns [39].

Due to the limited availability of annotated medical images, transfer learning has become a widely adopted strategy in CXR-based lung-disease classification. Instead of training deep models from scratch, many studies initialize CNN backbones with weights learned from large-scale natural-image datasets and subsequently fine-tune them on medical imaging tasks. Rahman et al. [40], Liang et al. [41], Ibrahim et al. [42], and Zubair et al. [43] investigated this direction for pneumonia and related chest-disease classification, demonstrating that pre-trained CNNs can provide effective visual representations when task-specific training data are limited.

These studies reflect the broader transition from handcrafted feature-based pipelines to data-driven representation learning in thoracic image analysis. By learning discriminative features directly from CXR images, CNN-based approaches have improved classification performance across several lung-disease detection tasks. Nevertheless, their effectiveness still depends on the availability and quality of labeled data, and most existing transfer-learning methods remain image-centered. They therefore provide limited ability to incorporate non-imaging clinical context, such as occupational exposure history or patient-profile information, which is particularly important for silicosis screening.

Building on the advancements in transfer learning for chest X-ray analysis, recent studies have further expanded the application of deep learning to other thoracic conditions, such as silicosis, a lung disease caused by prolonged exposure to silica dust. For instance, Tan et al. developed a deep learning framework for automated screening and staging of engineered-stone silicosis using chest X-rays, leveraging transfer learning from pre-trained convolutional neural networks to enhance detection accuracy despite limited annotated datasets [44]. Similarly, Li et al. proposed a deep learning approach for detecting silicosis from computed tomography images, employing a fine-tuned CNN model pre-trained on large-scale image datasets to improve feature extraction and classification performance [45]. These works highlight the adaptability of transfer learning in addressing specialized medical imaging tasks, further demonstrating the potential of CNNs to improve diagnostic precision and scalability in clinical applications.

While convolutional neural networks have long dominated chest X-ray analysis, recent research trends have shifted toward architectures that enhance both efficiency and interpretability. Instead of relying solely on deep convolutional backbones, new models integrate advanced scaling techniques and attention mechanisms to better capture complex thoracic patterns. For instance, [46] introduced a pneumonia detection framework using the EfficientNetV2L model, demonstrating that compound scaling across network depth, width, and resolution can significantly improve diagnostic accuracy while maintaining computational efficiency.

Beyond convolutional networks, transformer-based architectures have emerged as a powerful alternative for visual representation learning. [47] proposed an explainable hybrid transformer capable of multi-classifying various lung diseases from chest radiographs, illustrating how self-attention can uncover clinically meaningful spatial dependencies. Similarly, the MaxViT model developed by [48] combines both local windowed and global grid-based attention, effectively capturing multi-scale contextual information with improved parameter efficiency.

Although prior studies have demonstrated the value of deep learning for chest radiograph analysis and a small number of recent works have addressed silicosis-specific imaging tasks, existing approaches remain largely image-only and classifier-oriented. At the same time, broader vision-language research has shown the promise of cross-modal representation learning, but its application to silicosis screening with structured patient-profile data remains limited. In particular, we did not identify prior work that jointly models chest radiographs and structured occupational/clinical profiles within a retrieval-based, majority-voting framework. This gap motivates the proposed SiCLIP approach.

These advances signal a broader movement toward models that integrate global context awareness, explainability, and adaptability to limited medical datasets. Motivated by these developments, our proposed SiCLIP architecture explores a vision-language contrastive learning paradigm specifically tailored for silicosis detection. Unlike conventional CNN or transformer-based classifiers, SiCLIP leverages cross-modal representation alignment to enhance feature discrimination between diseased and healthy lung patterns, thereby improving performance under data-scarce clinical conditions.

3. Benchmark dataset

A central contribution of this study is the Silicosis Diagnosis Dataset (SDD), a multimodal dataset collected in Vietnam from workers occupationally exposed to silica dust. Unlike image-only datasets, SDD combines chest radiographs with structured worker-profile information, including occupational exposure duration, harmful habits, and clinically relevant symptoms. This combination makes SDD particularly valuable for studying multimodal silicosis screening in an under-represented real-world occupational-health setting.

3.1. Data collection

The SDD cohort was constructed from 1,633 workers with documented occupational exposure to silica dust in several industrial regions of Viet Nam. The participants were recruited from locations representing different silica-related working environments, including Thai Nguyen for metalworking, Hai Duong for cement production, Binh Dinh and Phu Yen for granite mining, and Dong Nai for brick manufacturing.

Data collection was conducted through field surveys and medical assessment performed by the research team. For each participant, demographic characteristics, occupational history, harmful habits, and clinically relevant symptoms were recorded to form the structured worker profile, as summarized in Table 2. Participants were eligible for inclusion if they had at least one year of occupational exposure at the time of data collection. Silicosis labels were assigned based on chest radiographs with a film size of 35×43 cm, interpreted according to the International Labour Organization (ILO) 2000 classification standards for pneumoconiosis, as shown in Fig. 4.

In the final cohort, 462 workers were diagnosed with silicosis. Fig. 1(a) presents the distribution of silicosis cases by gender and sampling location across the five provinces. The larger number of male cases reflects the predominance of men in the included high-risk occupations. Among the study sites, Thai Nguyen contributed the highest number of diagnosed cases. The distribution by years of employment in Fig. 1(b) indicates that silicosis cases tend to be more frequent among workers with longer occupational exposure. The relatively smaller number of cases in the 15–20 and 25+ year groups may reflect the smaller

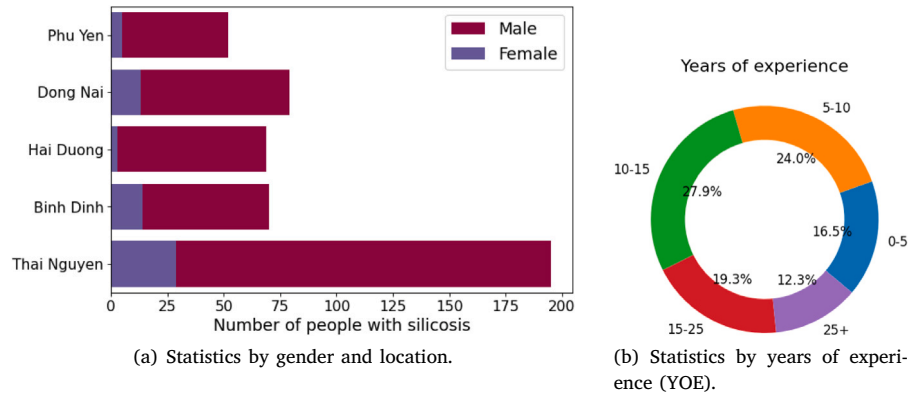


Fig. 1. Statistics of individuals diagnosed with silicosis.

number of workers with long-term employment histories in the sampled population.

Further analysis of the dataset, as illustrated in Fig. 2, reveals the distribution of occupational exposure years stratified by diagnostic status. This visualization highlights a clear divergence between individuals diagnosed with silicosis and those classified as normal. Specifically, workers without radiographic abnormalities are predominantly concentrated within the lower to moderate exposure range, reflecting the larger proportion of laborers with relatively shorter employment duration. In contrast, silicosis cases are disproportionately represented in the higher exposure intervals, with a noticeable clustering among individuals who have accumulated more than two decades of work in high-risk environments. This pattern reinforces the well-established dose–response relationship between cumulative silica exposure and disease manifestation. The longer a laborer remains engaged in occupations involving silica dust—such as metalworking, granite mining, brick manufacturing, or cement production—the more likely pathogenic changes are to develop and become detectable through standardized ILO 2000 radiographic assessment.

A complementary interpretation emerges when examining the four categorical features in Fig. 3, namely smoking status, difficulty breathing, respiratory history, and surgical history. According to the chi-square dependency analysis presented in Fig. 7, three of these medical features—difficulty breathing, respiratory history, and surgical history—do not exhibit a statistically significant association with diagnostic labels. Their distributions display a consistent decreasing trend from “no” to “yes” simply because the “yes” category is substantially less frequent in the population. In other words, their apparent differences largely reflect base-rate imbalance rather than a meaningful diagnostic signal.

In contrast, smoking status shows a markedly different pattern that aligns more closely with occupational epidemiology. While the normal group follows the same decreasing pattern, the silicosis group exhibits an almost equal number of smokers and non-smokers, deviating from the baseline trend observed in the other three features. This balance between “yes” and “no” within the silicosis subset suggests a stronger association between smoking and disease status than would be expected from marginal frequencies alone. Together with the pronounced separation observed in occupational exposure years, these findings indicate that—among the medical and lifestyle factors considered—occupational years and smoking are the two categorical variables demonstrating the clearest relationship with silicosis diagnosis, consistent with both the chi-square statistical analysis and established occupational-lung-health literature.

3.2. Data processing

The radiographic data were initially stored as DICOM (.dcm) files, a standard format for medical images. Such files can be opened using

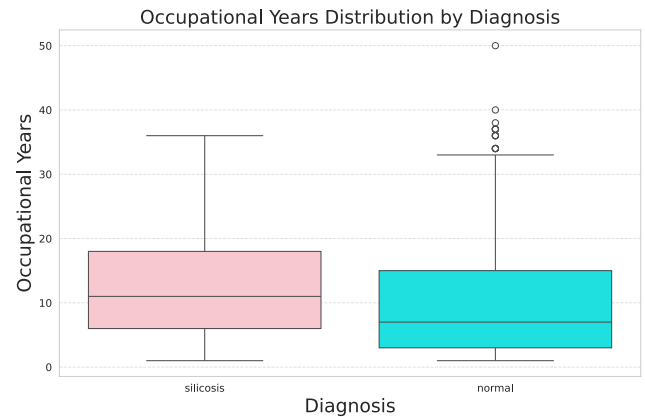


Fig. 2. Distribution of occupational exposure years by diagnosis category, comparing individuals diagnosed with silicosis and those with normal findings.

specialized software, including MicroDicom² and RadiAnt.³ However, manually reviewing and exporting the images to common formats such as .jpg or .png is impractical for a dataset of this size. To ensure a consistent and reproducible preprocessing workflow, we used Python scripts to automatically convert the DICOM images into .jpg files before model training and evaluation. We note that the conversion from the original DICOM format to JPG may reduce the fidelity of subtle radiographic information, including fine parenchymal patterns and low-contrast lesions that may be relevant for early silicosis detection.

The resultant dataset consists of a total of 1633 chest X-ray images, of which 462 depict silicosis lung disease and 1171 correspond to normal cases without the disease. The final analytic cohort used in this study was derived from a larger national screening pool through a structured sequence of technical-quality assessment, ILO-based radiographic classification, and case-level eligibility filtering. A detailed CONSORT-style accounting of this cohort-selection process, including the progression from 8086 screened radiographs to the final AI dataset and the reasons for exclusion at each stage, is provided in Supplementary Table 2.

In addition, worker profile data were stored in .dta format and subsequently transformed into textual records to conform to the input structure of our method. For each record, field names were concatenated with their respective values using a space, and the resulting components were separated by commas.

² MicroDicom software website: <https://www.microdicom.com/>

³ RadiAnt software website: <https://www.radiantviewer.com/>

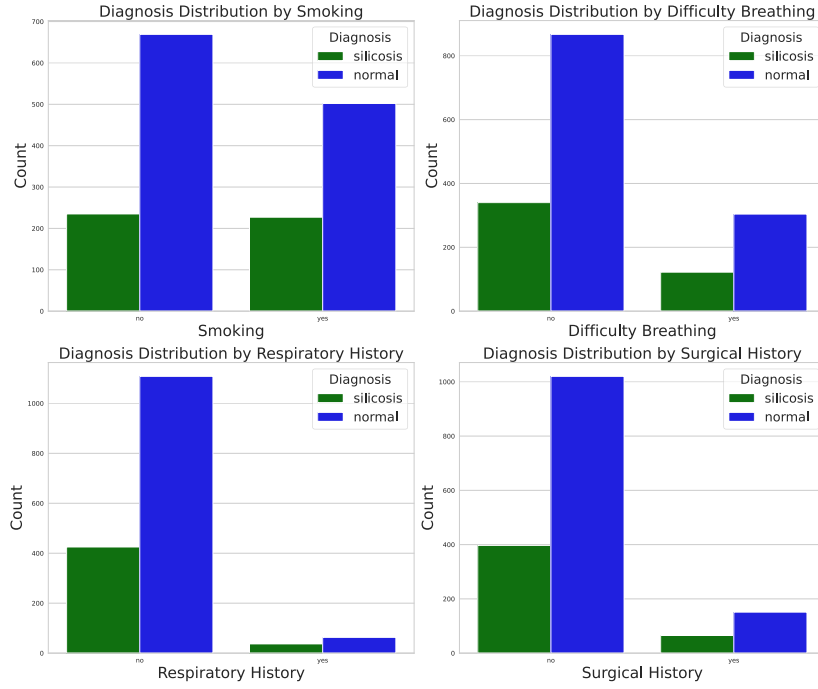


Fig. 3. Top-left: Distribution of smoking status across silicosis and normal patients. Top-right: Distribution of difficulty-breathing symptoms across the two diagnostic groups. Bottom-left: Distribution of respiratory-history status. Bottom-right: Distribution of surgical-history status. All four categorical features take only two discrete values (yes/no), and the vertical axis represents the count of patients in each category. Green bars correspond to silicosis, and blue bars correspond to normal cases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Reference class distribution of normal and silicosis cases used for stratified fold construction in the cross-validation protocol.

Subset	Normal	Silicosis	Total
Train	956	372	1328
Test	215	90	305
Total	1171	462	1633

For the purpose of model development and evaluation, we used a K -fold cross-validation protocol at the worker level. The class distribution previously reported as the train/test composition in Table 1 was retained as the reference distribution for stratified fold generation, ensuring that each fold followed, as closely as possible, the same balance between normal and silicosis cases. Thus, Table 1 should be interpreted as the reference class composition used to guide fold construction rather than as the sole final evaluation split. In each fold, one partition was used for evaluation and the remaining partitions were used for training and fold-specific model selection. Each worker contributed exactly one radiograph–profile pair and appeared in only one fold, which prevented information leakage across folds.

Data partitioning and leakage prevention. In the finalized AI dataset, each sample corresponded to one unique worker and was represented by one chest radiograph paired with one structured patient-profile record. Accordingly, dataset partitioning was performed at the worker level, such that no worker contributed data to more than one split. This design prevented information leakage across the training and test partitions.

4. Methodology

This section presents SiCLIP, our novel framework for diagnosing silicosis using multimodal data comprising Chest X-ray (CXR) images and patient profiles, as illustrated in Fig. 5. The core methodological

contribution of SiCLIP is a multimodal retrieval-and-voting framework that integrates chest radiographs and structured patient profiles by aligning them in a shared embedding space and aggregating diagnostic evidence across four retrieval streams. Our approach leverages the Contrastive Language-Image Pretraining (CLIP) model with a Vision Transformer backbone (CLIP-ViT) [49]. The methodology unfolds in four key stages, detailed in the subsequent subsections: We begin with the **Problem Definition**, outlining the task setup. Next, we describe **Feature Selection Using Chi-Square Test** to identify the most relevant patient profile attributes. We then detail the process of **Learning Multimodal Embeddings via CLIP Fine-tuning** to create a shared representation space. Finally, we explain the mechanism for **Interpretable Diagnosis through Multimodal Retrieval and Voting**, which utilizes vector databases and neighbor aggregation for prediction and explanation.

4.1. Problem definition

The diagnosis of silicosis presents a multimodal challenge, requiring the integrated analysis of visual evidence from Chest X-ray (CXR) images and pertinent textual information extracted from patient profiles, which typically include clinical symptoms, occupational exposure history, and relevant lifestyle habits. To address this, we formally define our dataset as $D = \{(x_i, t_i, y_i)\}_{i=1}^N$, consisting of N patient samples. For each sample i , $x_i \in \mathbb{R}^{H \times W \times 3}$ denotes the CXR image with height H , width W , and 3 color channels; t_i represents the corresponding textual patient profile; and $y_i \in \{0, 1\}$ is the binary ground truth label, where $y_i = 1$ indicates the presence of silicosis and $y_i = 0$ signifies its absence. The core objective of this research is to develop a model, parameterized by θ , encompassing image ($\mathcal{E}_I$) and text ($\mathcal{E}_T$) encoders, that effectively maps the visual features from x_i and the semantic information from t_i into a common, shared embedding space. Let $z_i^I = \mathcal{E}_I(x_i; \theta)$ and $z_i^T = \mathcal{E}_T(t_i; \theta)$ be the respective embeddings. Such a unified representation is intended to enable both accurate classification of silicosis status and facilitate the explanation of classification outcomes by integrating evidence from both imaging and textual modalities, thereby enhancing

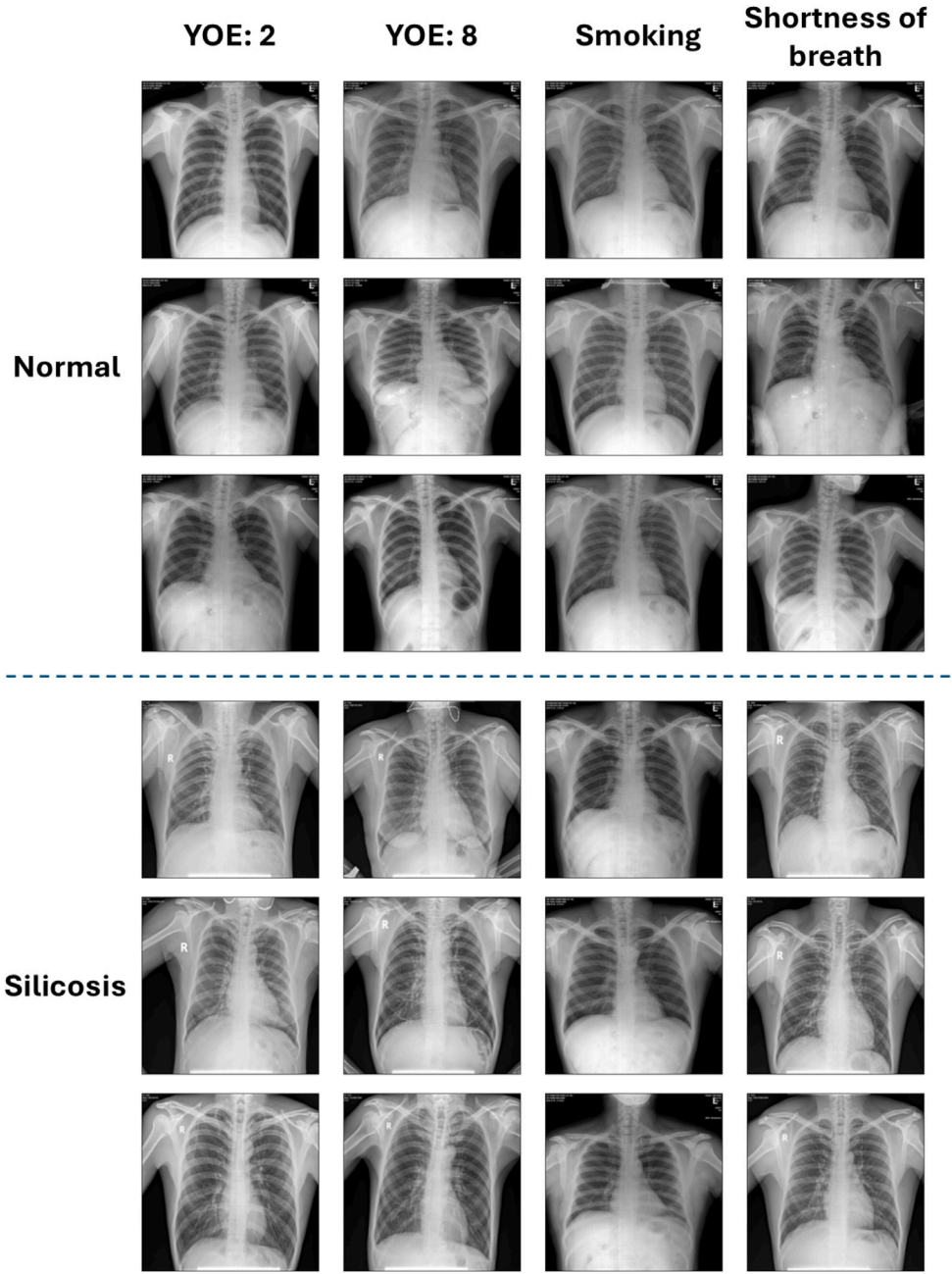


Fig. 4. Chest X-ray image samples from the dataset, including patient profiles with Years of Experience (YOE), bad habits such as smoking, symptoms like shortness of breath, and silicosis diagnosis.

model interpretability and clinical trust. Importantly, SiCLIP does not rely solely on a direct classifier over the joint representation. Instead, the shared space supports retrieval of similar cases across and within modalities, and the final diagnosis is derived by majority voting over retrieved evidence.

4.2. Feature selection using Chi-Square test

Our dataset D consists of tuples (x_i, t_i, y_i) , where t_i represents the textual patient profile corresponding to image x_i and label y_i . While t_i is treated as a single textual input for our text encoder $\mathcal{E}_T$, it is fundamentally derived from a collection of structured clinical, occupational, and demographic attributes associated with the patient. A comprehensive list of these potential source attributes is provided in Table 2. Not all attributes in Table 2 may be equally relevant or informative for the

specific task of silicosis diagnosis ($y_i \in \{0, 1\}$). Including irrelevant or noisy attributes in the construction of t_i could potentially hinder the performance of the text encoder $\mathcal{E}_T$. Therefore, to construct a more focused and effective textual profile t_i , we perform a feature selection step aimed at identifying which of these source attributes exhibit the strongest statistical association with the target diagnosis label y_i .

To quantify this association, we employ the Chi-Square (χ^2) test [50]. This test assesses the independence between two categorical variables. In our case, we evaluate the independence between each individual attribute (from Table 2) and the binary class label y_i . Prior to applying the test, numerical attributes (e.g., OccupationalYears, Weight) are appropriately discretized into categorical bins. For each attribute, a contingency table is constructed, tabulating the observed frequencies (O_{jk}) of each attribute value (where j runs from 1 to m , the number of distinct values for that attribute) for each class label ($k \in \{0, 1\}$), as conceptually shown in Table 3.

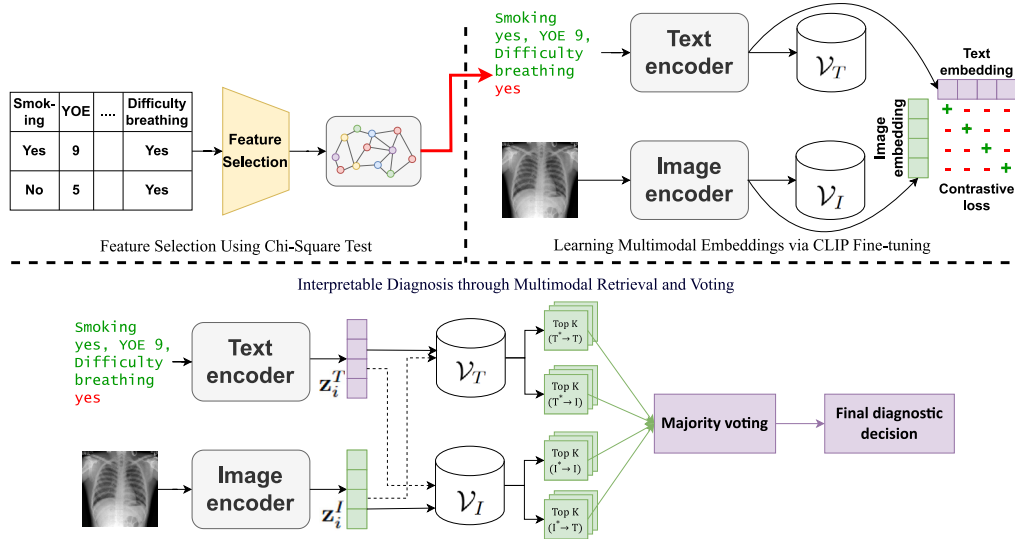


Fig. 5. The figure illustrates the SiCLIP architecture for multimodal silicosis diagnosis: Initially, Chi-Square feature selection processes patient profiles to generate concise text input. Subsequently, CLIP-ViT encoders handle the text and Chest X-ray (CXR) images, trained via Contrastive Learning to create a shared embedding space. Finally, during inference, a new case's embeddings are used for Top-K retrieval of similar cases from vector databases ($\mathcal{V}_T$, $\mathcal{V}_I$) across four streams ($T^* \rightarrow T$, $T^* \rightarrow I$, $I^* \rightarrow I$, $I^* \rightarrow T$); Majority Voting on the retrieved labels yields the final, interpretable Diagnostic Decision.

Table 2
Health and Occupational Data Attributes in SDD Patient Profiles.

Attribute	Description
OccupationalYears	Years of occupational experience in silica-exposed environment
Height	Height (cm)
Weight	Weight (kg)
SystolicBP	Systolic blood pressure (mmHg)
DiastolicBP	Diastolic blood pressure (mmHg)
HeartRate	Heart rate (beats per minute)
PhysicalFitness	Self-reported or assessed physical fitness level
SmokingStatus	Current or past smoking status (Yes/No/Ex-smoker)
CigarettesPerDay	Average number of cigarettes smoked per day (if smoker)
RespiratoryHistory	History of other respiratory diseases (e.g., asthma, TB) (Yes/No)
InternalMedicalHistory	History of significant internal medical conditions (Yes/No)
SurgicalHistory	History of major surgical procedures (Yes/No)
OccupationalDisease	History of other diagnosed occupational diseases (Yes/No)
OccDiseaseDetail	Specific occupational disease (if 'Yes' above)
SputumProduction	Presence of regular sputum production (Yes/No)
SputumType	Typical color/nature of sputum (e.g., clear, yellow, purulent, bloody)
SputumFrequency	Frequency or duration of sputum production
ShortnessOfBreath	Presence of shortness of breath (dyspnea), especially on exertion (Yes/No)
ChestPain	Presence of chest pain (Yes/No)
ChestPainLocation	Typical location of chest pain
ChestPainFrequency	Frequency of chest pain episodes
ChestPainNature	Character of chest pain (e.g., sharp, dull, tight)
WeightLossAmount	Unintentional weight loss and amount (if any)
ChestTightness	Sensation of chest tightness (Yes/No)
Fatigue	Presence of unusual fatigue or tiredness (Yes/No)
RunnyNose	Presence of chronic runny nose or sinus issues (Yes/No)
Label	Final diagnosis based on ILO criteria (1: Silicosis, 0: Normal)

Table 3

Conceptual contingency table structure. O_{jk} is the observed frequency for the j th value of the attribute in the k th class. R_j is the total count for the j th value.

Attribute value (j)	Class label (k)		Row total (R_j)
	Silicosis (1)	Normal (0)	
Value ₁ ($j=1$)	O_{11}	O_{10}	R_1
Value ₂ ($j=2$)	O_{21}	O_{20}	R_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$
Value _{m} ($j=m$)	O_{m1}	O_{m0}	R_m
Column Total (C_k)	C_1	C_0	Total (N)

The expected frequency (E_{jk}) for each cell under the null hypothesis of independence is calculated as $E_{jk} = (R_j \times C_k)/N$. The overall Chi-Square statistic for the feature is then computed by summing the squared differences between observed and expected frequencies, normalized by the expected frequency, across all cells:

$$\chi^2 = \sum_{k=0}^1 \sum_{j=1}^m \frac{(O_{jk} - E_{jk})^2}{E_{jk}} \quad (1)$$

A higher χ^2 score indicates a stronger dependency between the feature and the class label, suggesting greater relevance for classification. We compute this score for all candidate attributes detailed in Table 2. Based on these scores, we select the top- p attributes exhibiting the strongest association with the silicosis diagnosis. These selected, high-relevance features are then used to construct the concise and

informative textual patient profile (t_i) that serves as input to the text encoder component of our SiCLIP framework, as described in the following sections.

4.3. Learning multimodal embeddings via CLIP fine-tuning

Central to the SiCLIP method is the construction of a shared embedding space wherein image features derived from chest X-rays (CXR) and semantic information extracted from patient records are learned and effectively aligned. To achieve this, we leverage the capabilities of the Contrastive Language–Image Pretraining (CLIP) model [49], specifically utilizing a variant that employs a Vision Transformer (ViT) as the image encoder ($\mathcal{E}_I$) and a Transformer-based architecture as the text encoder ($\mathcal{E}_T$). Recognizing the substantial data requirements for training such models from scratch, we adopt a fine-tuning strategy applied to a CLIP model pre-trained on a large-scale, diverse dataset. This pre-training endows both encoders with a foundational, generalizable understanding of complex visual and linguistic concepts, thereby facilitating efficient adaptation to the specialized medical domain of silicosis diagnosis.

The fine-tuning process is specifically designed to adapt the pre-trained parameters of $\mathcal{E}_I$ and $\mathcal{E}_T$ to the nuances of the silicosis diagnosis task. This adaptation is performed using our curated Silicosis Diagnosis Dataset (SDD) training split, denoted as $\mathcal{D}_{\text{train}} = \{(x_i, t_i, y_i)\}_{i=1}^{N_{\text{train}}}$, which comprises N_{train} paired samples of CXR images (x_i) and their corresponding structured patient profile descriptions (t_i). The primary objective of this fine-tuning phase is to refine the model parameters such that the learned representations capture domain-specific features and relationships effectively. The fine-tuning procedure unfolds as follows:

- Batch processing:** The training data $\mathcal{D}_{\text{train}}$ is processed in mini-batches, each containing B image–text pairs (x_i, t_i) .
- Multimodal embedding generation:** For every image–text pair (x_i, t_i) within a mini-batch, the image encoder $\mathcal{E}_I$ maps the input image x_i to a D -dimensional image embedding vector $\mathbf{z}_i^I = \mathcal{E}_I(x_i)$, where $\mathbf{z}_i^I \in \mathbb{R}^D$. Concurrently, the text encoder $\mathcal{E}_T$ transforms the corresponding textual profile t_i into a text embedding vector $\mathbf{z}_i^T = \mathcal{E}_T(t_i)$ within the same D -dimensional space, $\mathbf{z}_i^T \in \mathbb{R}^D$.
- Embedding normalization:** A critical component of the CLIP framework is the L2 normalization applied to the output embeddings. Both image and text embeddings are projected onto the unit hypersphere:

$$\hat{\mathbf{z}}_i^I = \frac{\mathbf{z}_i^I}{\|\mathbf{z}_i^I\|_2}, \quad \hat{\mathbf{z}}_i^T = \frac{\mathbf{z}_i^T}{\|\mathbf{z}_i^T\|_2} \quad (2)$$

This normalization ensures that $\|\hat{\mathbf{z}}_i^I\|_2 = 1$ and $\|\hat{\mathbf{z}}_i^T\|_2 = 1$. By decoupling the magnitude from the direction, the model focuses on the angular relationship between embeddings. Consequently, the cosine similarity between normalized vectors simplifies to their dot product ($\hat{\mathbf{z}}_i^I \cdot \hat{\mathbf{z}}_j^T$), serving as an effective metric for semantic similarity in the shared space.

- Contrastive loss optimization:** The model parameters (within $\mathcal{E}_I$ and $\mathcal{E}_T$) are optimized by minimizing a symmetric contrastive loss function. This loss encourages high cosine similarity (dot product) between the normalized embeddings of corresponding image–text pairs ($\hat{\mathbf{z}}_i^I, \hat{\mathbf{z}}_i^T$) within the batch, while simultaneously pushing down the similarity between embeddings of non-corresponding pairs ($\hat{\mathbf{z}}_i^I, \hat{\mathbf{z}}_j^T$) where $i \neq j$. The symmetric loss L_{CLIP} is defined as the average of an image-to-text loss ($L_{I \rightarrow T}$) and a text-to-image loss ($L_{T \rightarrow I}$):

$$L_{\text{CLIP}} = \frac{1}{2} (L_{I \rightarrow T} + L_{T \rightarrow I}), \quad (3)$$

where, for a batch of size B ,

$$L_{I \rightarrow T} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\hat{\mathbf{z}}_i^I \cdot \hat{\mathbf{z}}_i^T / \tau)}{\sum_{j=1}^B \exp(\hat{\mathbf{z}}_i^I \cdot \hat{\mathbf{z}}_j^T / \tau)}, \quad (4)$$

$$L_{T \rightarrow I} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\hat{\mathbf{z}}_i^T \cdot \hat{\mathbf{z}}_i^I / \tau)}{\sum_{j=1}^B \exp(\hat{\mathbf{z}}_i^T \cdot \hat{\mathbf{z}}_j^I / \tau)}. \quad (5)$$

Here, τ represents a learnable temperature parameter that scales the logits before the softmax operation, controlling the sharpness of the distribution.

Upon completion of this fine-tuning process over a predefined number of epochs, the resulting optimized encoders, $\mathcal{E}_I$ and $\mathcal{E}_T$, are capable of generating robust, domain-adapted embeddings for novel CXR images and patient profiles pertinent to silicosis diagnosis. These embeddings effectively capture the specific features and relationships relevant to the task and are subsequently used to populate the vector databases utilized during the inference phase described in Section 4.4. Unlike standard CLIP adaptation for direct classification, our fine-tuning stage is explicitly optimized to support downstream multimodal retrieval, where both within-modality and cross-modality nearest-neighbor evidence are later aggregated for diagnosis.

4.4. Interpretable diagnosis through multimodal retrieval and voting

Following the creation of a unified embedding space via CLIP fine-tuning (Section 4.3), the SiCLIP diagnosis phase leverages this space for interpretable prediction. The process begins by populating vector databases, $\mathcal{V}_I$ and $\mathcal{V}_T$, with the pre-computed image ($\mathbf{z}_i^I$) and text ($\mathbf{z}_i^T$) embeddings, along with their corresponding ground truth labels (y_i), for all training samples. Efficient indexing (e.g., using FAISS [51]) allows rapid access to this knowledge base.

For a new query case (x^*, t^*), the fine-tuned encoders first generate its normalized embeddings, $\mathbf{z}^{I*} = \mathcal{E}_I(x^*)$ and $\mathbf{z}^{T*} = \mathcal{E}_T(t^*)$. Instead of direct classification, SiCLIP employs a retrieval-based approach to gather supporting evidence. These query embeddings initiate four parallel searches within the training databases to identify the top- k nearest neighbors, each reflecting a different similarity perspective:

- Image-to-Image ($I^* \rightarrow I$):** Query $\mathcal{V}_I$ with $\hat{\mathbf{z}}^{I*}$. Retrieve the set of labels $L_{I \leftarrow I^*}$ corresponding to the k nearest image embeddings in $\mathcal{V}_I$.

$$L_{I \leftarrow I^*} = \{y_j \mid (\hat{\mathbf{z}}_j^I, y_j, j) \in \text{Top-}k \text{ neighbors of } \hat{\mathbf{z}}^{I*} \text{ in } \mathcal{V}_I\}$$

- Image-to-Text ($I^* \rightarrow T$):** Query $\mathcal{V}_T$ with $\hat{\mathbf{z}}^{I*}$. Retrieve the set of labels $L_{T \leftarrow I^*}$ corresponding to the k nearest text embeddings in $\mathcal{V}_T$.

$$L_{T \leftarrow I^*} = \{y_j \mid (\hat{\mathbf{z}}_j^T, y_j, j) \in \text{Top-}k \text{ neighbors of } \hat{\mathbf{z}}^{I*} \text{ in } \mathcal{V}_T\}$$

- Text-to-Image ($T^* \rightarrow I$):** Query $\mathcal{V}_I$ with $\hat{\mathbf{z}}^{T*}$. Retrieve the set of labels $L_{I \leftarrow T^*}$ corresponding to the k nearest image embeddings in $\mathcal{V}_I$.

$$L_{I \leftarrow T^*} = \{y_j \mid (\hat{\mathbf{z}}_j^I, y_j, j) \in \text{Top-}k \text{ neighbors of } \hat{\mathbf{z}}^{T*} \text{ in } \mathcal{V}_I\}$$

- Text-to-Text ($T^* \rightarrow T$):** Query $\mathcal{V}_T$ with $\hat{\mathbf{z}}^{T*}$. Retrieve the set of labels $L_{T \leftarrow T^*}$ corresponding to the k nearest text embeddings in $\mathcal{V}_T$.

$$L_{T \leftarrow T^*} = \{y_j \mid (\hat{\mathbf{z}}_j^T, y_j, j) \in \text{Top-}k \text{ neighbors of } \hat{\mathbf{z}}^{T*} \text{ in } \mathcal{V}_T\}$$

Collectively, these searches retrieve $4k$ relevant exemplars from the training set. The final diagnostic prediction $\hat{y}^*$ for the query case (x^*, t^*) is determined by aggregating the labels collected from all four retrieval streams. Let the combined set of retrieved labels be $L_{\text{total}} = L_{I \leftarrow I^*} \cup L_{T \leftarrow I^*} \cup L_{I \leftarrow T^*} \cup L_{T \leftarrow T^*}$. This set contains up to $4k$ labels (potentially

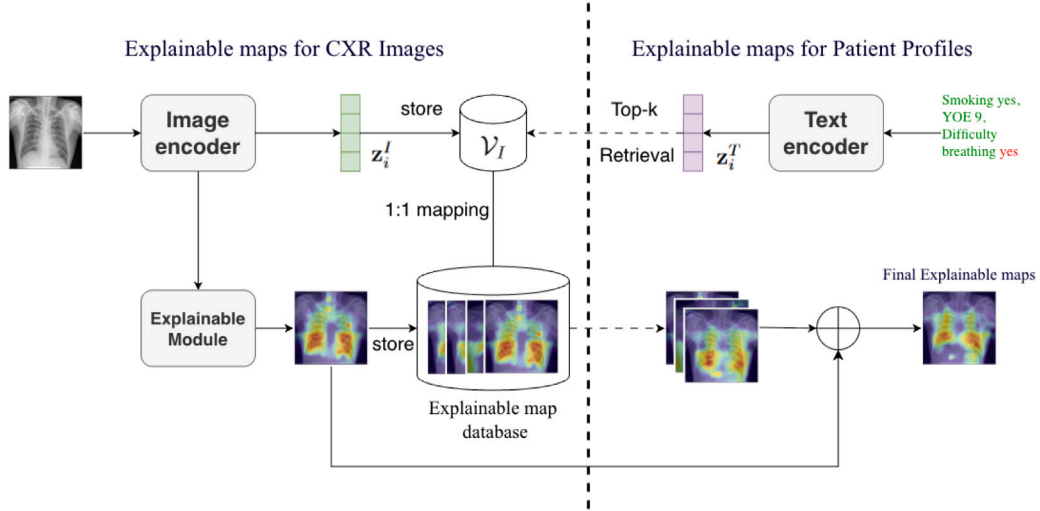


Fig. 6. The overall procedure of two processes: **Explainable map generation for CXR Images** and **Explainable map retrieval for Patient Profiles**. In the *Explainable map generation*, chest X-ray images are encoded into vector embeddings z_i^I , stored in a vector database V_I . Feature and gradient tensors from the encoder are processed by an explainability module to create explainable maps, stored in a dedicated database with a 1:1 mapping to V_I . In the *Explainable map retrieval*, a patient profile is encoded into a text embedding z_i^T , used for a top- k query in V_I to retrieve k relevant image embeddings and their corresponding explainable maps via the 1:1 mapping. These maps, along with the query CXR image's map (if provided), are aggregated to form the final explainable map.

fewer if duplicates exist across streams). The prediction is made via majority vote:

$$\hat{y}^* = \underset{c \in \{0,1\}}{\operatorname{argmax}} \sum_{y \in L_{\text{total}}} \mathbb{I}(y = c) \quad (6)$$

where $\mathbb{I}(\cdot)$ is the indicator function. This aggregation combines complementary evidence from radiograph-derived and profile-derived neighborhoods within the learned multimodal space. Because the final decision is based on retrieved exemplars, SiCLIP provides a transparent case-based reasoning pathway that can support inspection of the prediction process.

Upon confirmation of silicosis, our methodology elucidates critical regions within chest X-ray images that substantially influence the diagnostic determination, while seamlessly incorporating the effects of patients' behavioral patterns and clinical manifestations to underscore areas pivotal to accurate diagnostic conclusions. Fig. 6 depicts two key processes: the **Explainable map generation for CXR Images** and the **Explainable map retrieval for Patient Profiles**. In the former procedure, chest X-ray images are processed through an image encoder to produce vector embeddings z_i^I , which are stored in a vector database V_I . During this process, feature and gradient tensors extracted from the encoder are passed to an explainability module, generating explainable maps for each CXR image. These maps are stored in a dedicated explainable map database, with a one-to-one mapping to the corresponding entries in V_I . In the latter procedure, a patient profile is encoded via a text encoder to generate a text embedding z_i^T . This embedding is used to perform a top- k retrieval query in the vector database V_I , retrieving the k most relevant image embeddings. Based on the 1:1 mapping, the corresponding k explainable maps are retrieved from the explainable map database. Finally, these k maps, combined with the explainable map of a query CXR image (if provided), are aggregated to produce the final explainable map.

Explainable map for CXR Image: Given a CXR image, the CLIP-ViT model produces an image embedding vector z_i^I during inference. The explainable map for the image is generated as follows:

- Compute the sum of the embedding vector components:

$$s_i^I = \sum_{j=1}^d z_{i,j}^I \quad (7)$$

- Define the scalar s_i^I as the objective for backpropagation. Compute the gradient of s with respect to the model parameters by performing backpropagation:

$$\nabla_{\theta} s_i^I = \frac{\partial s_i^I}{\partial \theta} \quad (8)$$

where θ denotes the parameters of the CLIP-ViT model

This yields the gradient tensors $\{G_i^l\}$ for each layer l , where $G_i^l = \frac{\partial s_i^I}{\partial A_i^l}$, and A_i^l represents the activations at layer l . Additionally, retain the feature tensors $\{F_i^l\}$ computed during the forward pass.

- For each layer l , compute the layer-wise explainable map by element-wise multiplication of the gradient tensor and the feature tensor:

$$M_i^l = G_i^l \odot F_i^l \quad (9)$$

- Aggregate the explainable maps across all layers to obtain the final explainable map for the CXR image:

$$M_i^I = \sum_l M_i^l \quad (10)$$

- Store M_i^I in the explainability map database, which is integrated into the vector database V_I , alongside the corresponding image embedding z_i^I .

Explainable map for Patient Profile: During inference, a patient profile is provided as a text input, which is encoded into a text embedding vector z_i^T using the CLIP-ViT text encoder. To generate an explainable map for the patient profile, we leverage the top- k nearest image embeddings retrieved from the image vector database. The process is as follows:

- Query the image vector database using z_i^T to retrieve the top- k image embeddings $\{z_j^I\}_{j=1}^k$.
- For each retrieved image embedding z_j^I , fetch the corresponding explainable map M_j^I from the explainable map database.
- Compute a weighted combination of the top- k explainable maps, where the weight w_j is the similarity score (e.g., cosine similarity) between z_i^T and z_j^I , the explainable map for the patient profile is:

$$M_i^T = \sum_{j=1}^k w_j M_j^I, \quad \text{where } w_j = \frac{z_i^T \cdot z_j^I}{\|z_i^T\| \|z_j^I\|} \quad (11)$$

Final explainable map: To provide a comprehensive interpretation of the prediction of the model, the explainable maps for the CXR image and the patient profile are combined. Given the explainable map M_i^I for the queried CXR image and the explainable map M_i^T for the patient profile, the final explainable map is calculated as

$$M_i = M_i^I + M_i^T \quad (12)$$

This composite map M_i represents the combined contribution of the CXR image and the patient profile to the model's decision, enabling an interpretable visualization of the diagnostic process.

5. Experimental results

In this section, our experiments are conducted to address the following research questions (RQs):

- RQ1: How does SiCLIP's classification accuracy compare to baseline models on medical imaging datasets?
- RQ2: How does varying top-k neighbors in SiCLIP affect its classification accuracy and efficiency?
- RQ3: Which SiCLIP features are most significant for disease classification per the Chi-Square test?
- RQ4: How do SiCLIP's explainable visualization maps, conditioned on patient profiles, highlight differences in disease-specific patterns between silicosis and normal cases in medical images?
- RQ5: How does SiCLIP compare descriptively with expert radiologist consensus on a selected subset of cases?

5.1. Experimental settings

All experiments were conducted on hardware featuring an NVIDIA RTX 4090 GPU with 24 GB of RAM. Input chest X-ray images underwent standard preprocessing, notably resizing to a uniform 224×224 pixel resolution. Although this preprocessing step standardizes the input for model training, it may suppress fine-grained radiographic detail and therefore limit sensitivity to subtle lesion patterns. The derivation of the final analytic cohort from the full screened population is summarized in Supplementary Table 2. Instead of training from scratch, we adopted a fine-tuning approach for the CLIP-ViT encoders, leveraging powerful pre-trained weights. This fine-tuning process spanned 50 epochs. To enhance model robustness and mitigate overfitting, standard data augmentation techniques (random horizontal flips and rotations) were applied during the training phase. Model optimization was managed using the RAdam [52] optimizer. The inference stage, central to our SiCLIP method, utilized a top- k nearest neighbor retrieval strategy from the generated vector databases. Specific configuration details and hyperparameter values used throughout the training and inference processes are comprehensively listed in Table 4.

5.2. Evaluation metrics

Because the revised experimental protocol uses stratified K -fold cross-validation at the worker level, all evaluation metrics are computed on *held-out* data only, i.e., on the evaluation partition of each fold. Each worker contributes exactly one radiograph-profile pair and appears in the held-out portion of exactly one fold, thereby preventing information leakage across folds. Unless otherwise stated, we treat *Silicosis* as the positive class and *Normal* as the negative class.

Let TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively, computed from held-out predictions. To ensure consistency with all results reported in Tables 5–7, we evaluate each method using the following metrics.

Table 4

Summary of key hyperparameters and experimental settings.

Data preprocessing	
Input Image Size	224×224 pixels
Training Phase	
Training Type	Fine-tuning (CLIP-ViT)
Base Model	CLIP-ViT (ViT-L/14 @ 336px)
Epochs	50
Batch Size	32
Data Augmentation	RandomHorizontalFlip, RandomRotation
Optimizer	Rectified Adam (RAdam)
Initial Learning Rate	1×10^{-5}
Optimizer Epsilon (ϵ)	1×10^{-8} (default)
Learning Rate Scheduler	Constant LR
Weight Decay	0.01
Loss Function	Symmetric Contrastive Loss (CLIP Loss)
Temperature (τ)	Learnable (CLIP default)
Inference Phase	
Top- k Neighbors Tested	$k \in \{1, 3, 5\}$
Vector Database Index	FAISS
Similarity Metric	Cosine Similarity (L2 normalized dot)

Accuracy. Accuracy measures the proportion of correctly classified held-out cases:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \quad (13)$$

Precision and recall. For the Silicosis class, precision and recall are defined as

$$P_{\text{sil}} = \frac{TP}{TP + FP}, \quad R_{\text{sil}} = \frac{TP}{TP + FN}. \quad (14)$$

Here, R_{sil} is equivalent to *Sensitivity*.

For the Normal class, we analogously define

$$P_{\text{norm}} = \frac{TN}{TN + FN}, \quad R_{\text{norm}} = \frac{TN}{TN + FP}. \quad (15)$$

Here, R_{norm} is equivalent to *Specificity*.

Macro F1-score. Since the dataset is class-imbalanced, we report the macro F1-score, computed as the arithmetic mean of the per-class F1-scores:

$$F1_{\text{sil}} = \frac{2P_{\text{sil}}R_{\text{sil}}}{P_{\text{sil}} + R_{\text{sil}}}, \quad F1_{\text{norm}} = \frac{2P_{\text{norm}}R_{\text{norm}}}{P_{\text{norm}} + R_{\text{norm}}}, \quad (16)$$

$$\text{Macro-F1} = \frac{1}{2} (F1_{\text{sil}} + F1_{\text{norm}}). \quad (17)$$

In the remainder of the paper, we refer to this quantity simply as *F1 Score* for brevity.

Sensitivity and specificity. In addition to Accuracy and Macro-F1, we explicitly report the two clinically important class-conditional recall measures:

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP}. \quad (18)$$

These correspond to the recall of the Silicosis and Normal classes, respectively.

Class-wise AUC-ROC. To assess ranking performance independently of a fixed decision threshold, we also report one-vs-rest AUC-ROC values for each class. Specifically, AUC_{sil} is computed by treating Silicosis as the positive class and AUC_{norm} is computed by treating Normal as the positive class.

Cross-validation aggregation. For each fold, all metrics are computed on the held-out partition only. After completing all K folds, we pool the out-of-fold predictions from all held-out workers and use these pooled patient-level predictions to summarize final performance. This pooled evaluation protocol is used consistently for all metrics reported in Tables 5–7.

Table 5

Performance comparison against baseline models on the silicosis classification task. SiCLIP results are reported for the optimal configuration ($k = 3$, multimodal). Best results are in bold. Numbers within parentheses indicate 95% CI.

Model	Params	Image	Text	Accuracy	F1 Score
MobileNet_V3_small	4.4M	✓		0.72 (0.67, 0.77)	0.65 (0.60, 0.70)
DenseNet201	18.3M	✓		0.85 (0.81, 0.89)	0.70 (0.65, 0.76)
Swin_V2_b	87M	✓		0.87 (0.83, 0.90)	0.68 (0.63, 0.74)
ConvNeXtLarge [56]	197.9M	✓		0.88 (0.84, 0.91)	0.72 (0.66, 0.78)
Qwen2-VL [57]	7B	✓		0.80 (0.75, 0.84)	0.60 (0.54, 0.66)
Qwen2-VL [57]	7B		✓	0.78 (0.73, 0.83)	0.58 (0.52, 0.65)
Qwen2-VL [57]	7B	✓	✓	0.86 (0.82, 0.89)	0.70 (0.65, 0.76)
SiCLIP (Ours, $k = 3$)	890M	✓	✓	0.94 (0.91, 0.97)	0.85 (0.81, 0.89)

Confidence intervals. All reported 95% confidence intervals are estimated from the *pooled patient-level held-out predictions* obtained under the worker-level stratified K -fold cross-validation protocol. For each model, we apply non-parametric bootstrap resampling at the *worker (patient)* level: at each bootstrap iteration, workers are sampled with replacement from the pooled out-of-fold prediction set, and the corresponding metric is recomputed on the resampled set. This procedure is applied to all reported performance metrics, including Accuracy, Macro-F1, Sensitivity, Specificity, and class-wise one-vs-rest AUC-ROC. The final 95% confidence interval is taken as the 2.5th and 97.5th percentiles of the bootstrap distribution.

5.3. Comparison with baseline approaches (RQ1)

We evaluated SiCLIP against a range of baseline models on the SDD dataset for the silicosis classification task. The baselines include several well-established image-only CNN architectures (MobileNetV3-small [53], DenseNet201 [54], SwinV2-b [55], ConvNeXtLarge [56]) and a recent large vision–language model (Qwen2-VL [57]). For a fair comparison, SiCLIP is evaluated using its optimal configuration identified in Section 5.4 ($k = 3$, multimodal input).

Tables 5 and 6 summarize the quantitative results, while Table 7 provides a detailed view of the corresponding confusion-matrix summaries with 95% confidence intervals. Within the reported internal evaluation setting, SiCLIP achieved the best performance among the evaluated baselines across the reported metrics. SiCLIP consistently outperforms all baselines across multiple evaluation metrics. In Table 5, SiCLIP achieves the highest overall Accuracy (0.94, 95% CI: 0.90–0.97) and F1 Score (0.85, 95% CI: 0.81–0.89), surpassing ConvNeXtLarge—the strongest image-only baseline—with notable margins. The improvement in F1 Score underscores SiCLIP’s enhanced balance between precision and recall, effectively mitigating the class imbalance inherent to the dataset.

From the confusion matrix analysis (Table 7), SiCLIP correctly identifies 82 out of 90 silicosis cases on average, corresponding to a Sensitivity of 0.91 (95% CI: 0.86–0.96), while maintaining a low number of false positives (15 out of 215 normal cases), yielding a Specificity of 0.93 (95% CI: 0.89–0.97). These results demonstrate the model’s strong ability to detect true silicosis cases without compromising reliability in normal predictions. Moreover, SiCLIP achieves near-perfect AUC ROC values for both silicosis (0.99, 95% CI: 0.98–1.00) and normal (0.999, 95% CI: 0.997–1.000) classes, indicating a highly discriminative decision boundary between positive and negative cases.

In contrast, Qwen2-VL exhibits moderate performance in the multimodal setting (Accuracy 0.86, F1 0.70), but its performance drops substantially when restricted to a single modality—either image or text—reflecting its dependence on explicit cross-modal alignment. SiCLIP, by comparison, effectively leverages complementary diagnostic cues from chest X-ray images and patient clinical profiles through its CLIP-based joint embedding and retrieval mechanism. These results indicate that, within this internal evaluation setting, SiCLIP provided a more accurate and better-balanced diagnosis than the evaluated baselines.

5.4. Ablation study on the effect of top- k neighbors (RQ2)

To examine the effect of the number of retrieved neighbors (k) and the contribution of each input modality, we conducted an ablation study by varying $k \in 1, 3, 5$ and evaluating SiCLIP under three conditions: image-only retrieval (Image $\rightarrow$ Image DB, Text $\rightarrow$ Image DB), text-only retrieval (Image $\rightarrow$ Text DB, Text $\rightarrow$ Text DB), and combined multimodal retrieval (all four retrieval streams). The results are summarized in Table 8.

Several key observations can be drawn from the findings:

- **Impact of k :** Across all modality configurations, performance improves as k increases from 1 to 3, indicating that aggregating information from a small neighborhood of similar cases enhances the model’s robustness. For instance, under the multimodal setting, the Accuracy increases from 0.90 (95% CI: 0.86–0.93) at $k = 1$ to 0.94 (95% CI: 0.90–0.97) at $k = 3$, and the F1 Score rises from 0.78 (95% CI: 0.73–0.83) to 0.85 (95% CI: 0.81–0.89). However, expanding the neighborhood further to $k = 5$ slightly reduces performance (Accuracy 0.92, F1 Score 0.81), suggesting that including less relevant neighbors introduces noise. Therefore, $k = 3$ consistently yields the best overall results.
- **Modality contribution:** SiCLIP achieves strong results even when relying on a single modality. When only image features are used, the model attains 0.92 (95% CI: 0.88–0.95) Accuracy and 0.82 (95% CI: 0.77–0.87) F1 Score at $k = 3$. Similarly, the text-only configuration reaches 0.91 (95% CI: 0.87–0.94) Accuracy and 0.80 (95% CI: 0.75–0.85) F1 Score. These results demonstrate that both the CXR images and patient text profiles contain rich and complementary diagnostic information, and that SiCLIP can leverage either source effectively in isolation.
- **Multimodal synergy:** The highest performance is achieved when both modalities are combined, with the multimodal SiCLIP reaching 0.94 (95% CI: 0.90–0.97) Accuracy and 0.85 (95% CI: 0.81–0.89) F1 Score at $k = 3$. This clearly illustrates the synergistic effect of fusing image and text representations. By aligning both modalities in a shared embedding space, SiCLIP effectively captures cross-modal relationships and enhances diagnostic reliability.

5.5. Feature relevance analysis using Chi-Square test (RQ3)

To better understand the contribution of individual patient profile attributes to the silicosis diagnosis and to validate the feature selection process outlined in Section 4.2, we visualized the association strength between the most relevant selected features and the final diagnosis label. As described previously, the Chi-Square (χ^2) test was employed to quantify the statistical dependence between each categorical (or discretized numerical) attribute from Table 2 and the binary ‘Label’ (Silicosis vs. Normal). A higher χ^2 score indicates a stronger association, suggesting the feature is more discriminative for the classification task.

Table 6

Comparison of diagnostic performance metrics across baseline and multimodal models on the silicosis classification task. The table reports Sensitivity, Specificity, and class-wise AUC ROC (Silicosis and Normal). SiCLIP results are shown for the optimal configuration ($k = 3$, multimodal). Best-performing values are highlighted in bold. Numbers within parentheses denote 95% confidence intervals.

Model	Image	Text	Sensitivity	Specificity	AUC ROC	
					Silicosis	Normal
MobileNet_V3_small	✓		0.67 (0.57, 0.76)	0.74 (0.67, 0.81)	0.97 (0.94–0.99)	0.99 (0.97–1.00)
DenseNet201	✓		0.78 (0.70, 0.85)	0.87 (0.82, 0.91)	0.96 (0.94–0.98)	0.99 (0.98–1.00)
Swin_V2_b	✓		0.81 (0.74, 0.88)	0.88 (0.84, 0.92)	0.97 (0.95–0.99)	0.99 (0.98–1.00)
ConvNeXtLarge [56]	✓		0.83 (0.77, 0.89)	0.89 (0.85, 0.93)	0.98 (0.96–1.00)	0.99 (0.98–1.00)
Qwen2-VL [57]	✓		0.73 (0.65, 0.80)	0.77 (0.70, 0.83)	0.95 (0.91–0.97)	0.97 (0.94–0.99)
Qwen2-VL [57]		✓	0.71 (0.63, 0.78)	0.75 (0.68, 0.82)	0.94 (0.90–0.96)	0.96 (0.93–0.98)
Qwen2-VL [57]	✓	✓	0.81 (0.74, 0.88)	0.87 (0.82, 0.91)	0.96 (0.94–0.98)	0.98 (0.96–0.99)
SiCLIP (Ours, $k = 3$)	✓	✓	0.91 (0.86, 0.96)	0.93 (0.89, 0.97)	0.98 (0.95–0.99)	0.98 (0.97–0.99)

Table 7

Confusion-matrix summary expressed as cross-validated patient-level counts (or expected counts under the reference evaluation composition in Table 1), with 95% confidence intervals obtained by patient-level bootstrap resampling.

Model	TP	FN	FP	TN
MobileNetV3-small	60 (54–66)	30 (24–36)	56 (48–64)	159 (150–167)
DenseNet201	70 (63–76)	20 (14–27)	28 (21–36)	187 (179–195)
SwinV2-b	73 (66–79)	17 (11–24)	26 (19–33)	189 (182–197)
ConvNeXtLarge [56]	75 (68–81)	15 (9–22)	24 (18–31)	191 (184–199)
Qwen2-VL [57] (image-only)	66 (59–73)	24 (17–31)	50 (42–58)	165 (157–173)
Qwen2-VL [57] (text-only)	64 (57–71)	26 (19–33)	54 (46–62)	161 (153–169)
Qwen2-VL [57] (multi)	73 (66–79)	17 (11–24)	28 (21–36)	187 (179–195)
SiCLIP (Ours, $k = 3$)	82 (78–86)	8 (4–12)	15 (11–20)	200 (194–206)

Table 8

Ablation study on the effect of top- k neighbors and input modality on SiCLIP performance. Accuracy and F1 Score (with 95% confidence intervals) are reported. Best overall results are in bold. Numbers within parentheses denote 95% confidence intervals.

Top- k	Image Input Active	Text Input Active	Accuracy	F1 Score
$k = 1$	✓		0.88 (0.84, 0.92)	0.74 (0.68, 0.79)
		✓	0.87 (0.83, 0.91)	0.72 (0.66, 0.78)
	✓	✓	0.90 (0.86, 0.93)	0.78 (0.73, 0.83)
$k = 3$	✓		0.92 (0.88, 0.95)	0.82 (0.77, 0.87)
		✓	0.91 (0.87, 0.94)	0.80 (0.75, 0.85)
	✓	✓	0.94 (0.90, 0.97)	0.85 (0.81, 0.89)
$k = 5$	✓		0.90 (0.86, 0.93)	0.79 (0.74, 0.84)
		✓	0.89 (0.85, 0.92)	0.77 (0.72, 0.82)
	✓	✓	0.92 (0.88, 0.95)	0.81 (0.76, 0.86)

Fig. 7 presents a graph visualization of these relationships for a subset of the top-ranking features identified by the Chi-Square test. Each node represents either a patient profile attribute (e.g., ‘Occupational Years’, ‘Shortness Of Breath’) or the target ‘Label’. The edges connecting the attributes to the ‘Label’ node represent the direct association between that feature and the diagnosis. The numerical value displayed on each edge corresponds to the calculated Chi-Square score (potentially normalized or scaled for visualization purposes), reflecting the strength of that association. Edges between attribute nodes represent inter-feature associations, also quantified using a suitable measure like the Chi-Square score between those features.

From the analysis presented in Fig. 7, several key insights emerge:

- **Strongest predictors:** ‘Occupational Years’ (exposure duration) and ‘Medical History’ (likely referring to general or respiratory medical history) exhibit the strongest direct associations with the ‘Label’, showing Chi-Square scores of 0.63 and 0.64, respectively. This aligns perfectly with clinical understanding, as prolonged silica exposure is the primary cause of silicosis, and prior medical conditions can influence susceptibility or presentation.
- **Relevant symptoms and history:** ‘Shortness Of Breath’, a cardinal symptom of silicosis, shows a significant association score of 0.51 with the label. Similarly, ‘Internal Medical History’ (0.42) and ‘Surgical History’ (0.36) also demonstrate relevance, although to a lesser extent. These factors likely contribute contextual information important for diagnosis or differential diagnosis.

- **Inter-feature relationships:** The graph also reveals correlations between features, such as the link between ‘Medical History’ and ‘Occupational Years’ (0.58) or ‘Shortness Of Breath’ (0.63). This suggests potential multicollinearity but also highlights how different aspects of a patient’s profile are often interrelated.

This analysis supports the relevance of the features selected via the Chi-Square test for constructing the textual profile representation used by SiCLIP. In particular, features such as Occupational Years, Smoking, and Shortness of Breath show stronger statistical association with the binary diagnosis label in the present dataset. However, Fig. 7 should be interpreted only as a descriptive summary of feature–label association patterns. It does not constitute a quantitative validation of interpretability, explanation faithfulness, stability, or clinical reliability. Rather, it provides supportive context for understanding which non-imaging variables appear most informative under the current dataset and experimental setting.

5.6. Visual analysis (RQ4)

The GradCAM [58] visualization in Fig. 8 illustrates the influence of specific patient profile factors on chest X-ray images for two distinct cases: a normal lung and a lung affected by silicosis. The visualization is organized into a grid with rows and columns, each representing different aspects of the analysis. The first row displays the X-ray image

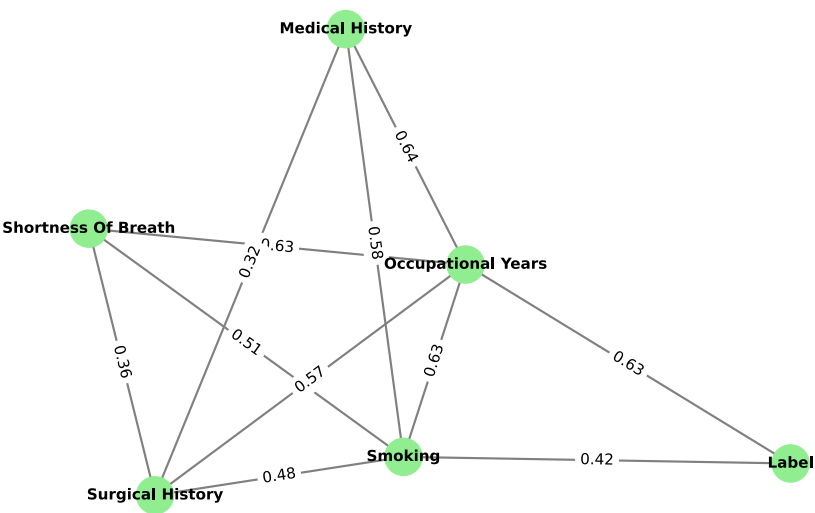


Fig. 7. Graph visualizing the association strength (based on Chi-Square scores) between selected patient profile features and the Silicosis diagnosis label. Higher edge weights indicate stronger association.

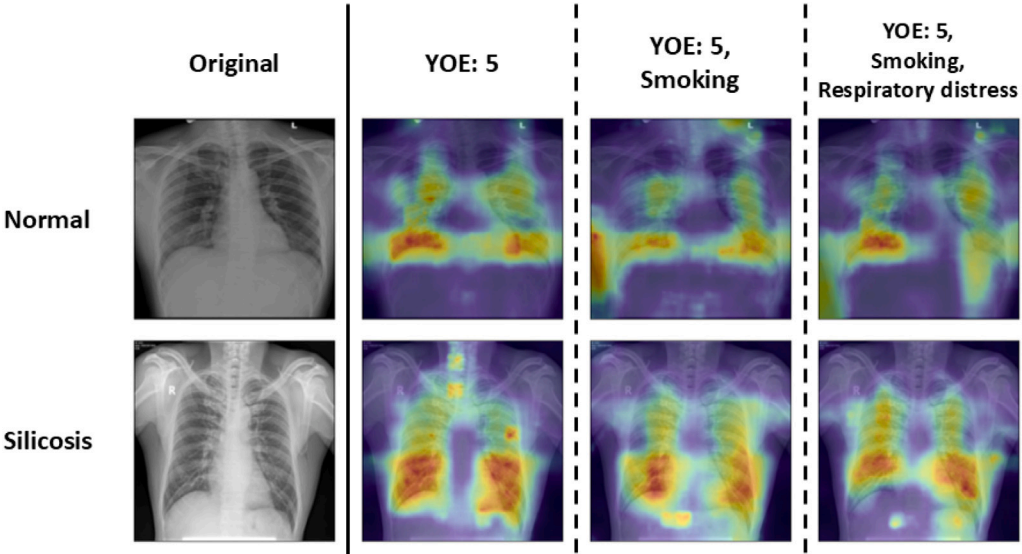


Fig. 8. Illustrative GradCAM-based qualitative visualization of the influence of retrieved patient-profile factors, including years of occupational exposure (YOE), smoking status, and respiratory symptoms.

of a healthy lung, labeled as “Normal.” The second row shows the X-ray image of a lung diagnosed with silicosis, a condition often caused by prolonged exposure to silica dust. The grid has four columns, each representing either the original X-ray or a GradCAM heatmap overlaid on the X-ray based on specific patient profile factors:

- The first column shows the original chest X-ray images for both the normal and silicosis cases, providing a baseline for comparison.
- The second column displays the GradCAM heatmap highlighting regions of the X-ray that are most influenced by the patient’s years of experience (YOE) in harmful environments, set at 5 years. For the normal case, the heatmap shows minimal activation, indicating low impact, while for the silicosis case, there are stronger activations in the lung regions, suggesting a correlation with prolonged exposure.
- The third column presents the GradCAM heatmap considering both 5 years of experience in harmful environments and a smoking history. In the normal case, the heatmap remains relatively

- unchanged, but in the silicosis case, the heatmap intensifies in the lower lung regions, indicating that smoking may exacerbate the effects of environmental exposure.
- The fourth column shows the GradCAM heatmap incorporating all three factors: 5 years of experience in harmful environments, smoking, and respiratory distress. For the normal case, the heatmap shows slight increases in activation, but for the silicosis case, the heatmap highlights extensive areas across the lungs, reflecting the combined impact of these risk factors on lung health.

The GradCAM heatmaps use a color gradient where warmer colors (yellow to red) indicate higher importance of the corresponding regions in the X-ray for the model’s decision-making process. In the normal case, the heatmaps show limited activation, suggesting that the model does not associate these risk factors with significant abnormalities. In contrast, the silicosis case exhibits progressively stronger activations as more risk factors are considered, particularly in the lower and central lung regions, which are commonly affected in silicosis due to the deposition of silica particles. This visualization underscores the cumulative

effect of environmental exposure, smoking, and respiratory distress in contributing to the severity of silicosis, providing valuable insights for medical diagnosis and risk assessment. We emphasize that this figure is intended only as an illustrative example of how the explainability mechanism behaves when conditioned on different combinations of profile factors. It should not be interpreted as formal evidence that interpretability has been validated.

5.7. Exploratory comparison with expert radiologists (RQ5)

The preceding experimental results suggest that SiCLIP may have practical value as a decision-support tool in occupational screening settings. However, we emphasize that the expert comparison presented in this section is *exploratory* and is intended only to provide descriptive context regarding the potential supportive role of the model. It is *not* designed as a statistically controlled reader study and should therefore not be interpreted as definitive evidence that the model is superior to expert radiologists in clinical practice.

Because expert review of chest radiographs is inherently time-intensive and requires sustained specialist attention, it was not feasible to conduct an independent human reading study over the entire 1633-case dataset used for model development and cross-validated evaluation. We therefore constructed a dedicated expert-review subset consisting of 417 cases (107 Silicosis and 310 Normal). This subset was selected to include the highest-quality and most representative radiographs available for reader assessment, while preserving a class distribution broadly consistent with that of the full dataset. The purpose of this design was to obtain a clinically meaningful and practically manageable sample for human evaluation, rather than to replicate the full-scale model evaluation protocol on all cases. These images were evaluated by the Council of the Occupational Radiology Association for Pneumoconiosis, affiliated with the Vietnamese Occupational Medicine Association in Thai Nguyen. The council, led by Prof. Dr. Do Van Ham, has extensive experience in occupational lung disease, including silicosis. The expert review process was conducted in two stages: (i) verification of X-ray implementation according to occupational-medicine technical standards, and (ii) detailed image assessment to determine worker health status. The council performed two independent rounds of review, after which a final consensus conclusion was issued by the Chairman of the Council.

Importantly, this comparison has several methodological limitations. First, the analysis is reported at the level of a consensus expert outcome, rather than as a multi-reader paired study with reader-specific performance distributions. Second, we do not report reader-level uncertainty estimates or inter-reader variability. Third, we do not perform a formal paired model-versus-reader statistical comparison on the same cases (e.g., a dedicated matched significance test). Accordingly, the present results should be interpreted as descriptive rather than confirmatory.

Expert performance and limitations in silicosis detection. The relatively low sensitivity observed in the expert assessment is consistent with prior clinical evidence showing that chest radiography may underdetect early or mild silicosis. For example, among workers exposed to artificial stone, chest X-ray achieved only 48% sensitivity (95% CI: 29–68) when benchmarked against high-resolution CT (HRCT) [59]. This limitation is particularly relevant for subtle parenchymal abnormalities and faint diffuse opacities, which may be difficult to identify even for experienced readers.

In addition, the comparison in our study should be interpreted in light of differences between the expert assessment setting and the reference labels used in the dataset. Ground-truth labels were assigned according to the ILO 2000 classification standard, whereas the participating radiologists assessed the images in a routine clinical reading framework. Differences in diagnostic criteria, as well as variation in image contrast, background noise, and acquisition settings, may contribute to disagreement between expert reads and dataset labels.

Table 9

Descriptive confusion-matrix summary for the expert consensus reading on the 417-image exploratory subset.

		Predicted class	
		Silicosis	Normal
Actual Class	Silicosis	40	67
	Normal	4	306

These factors may partly explain the lower sensitivity observed in the descriptive expert comparison.

Interpretation of the exploratory comparison. Table 9 summarizes the consensus expert reading on the selected subset. These results are useful for illustrating the difficulty of silicosis detection on chest radiographs within the present study setting and for motivating the potential role of AI-assisted support. However, because this analysis does not constitute a formally controlled paired reader study, we refrain from making a statistical superiority claim for SiCLIP relative to expert radiologists. A more rigorous comparison would require a dedicated same-case model-versus-reader evaluation, reader-level uncertainty estimation, and an appropriate formal statistical framework. We therefore position the current findings as preliminary evidence supporting the potential utility of SiCLIP as an assistive tool, rather than as a definitive comparative validation against human experts.

6. Conclusion and discussions

In this study, we introduced the Silicosis Diagnosis Dataset (SDD) and proposed SiCLIP, a multimodal retrieval framework that integrates chest X-ray images with structured patient-profile information for silicosis-related screening and classification support. Within the reported internal evaluation on SDD, SiCLIP achieved higher Accuracy and F1-score than several strong image-only deep learning baselines, and also outperformed the compared multimodal VLM baseline in our experimental setting.

A further strength of SiCLIP is that it can still operate when only patient-profile data are available. However, this result should be interpreted cautiously. The strong profile-only performance suggests that patient metadata contain substantial predictive signal, but it also raises the possibility that non-imaging features may dominate the prediction in some cases. We therefore view profile-only inference as a fallback setting rather than a replacement for multimodal assessment when chest radiographs are available.

SiCLIP also provides case-based interpretability support through retrieved similar cases and qualitative saliency visualizations. We emphasize that this evidence remains associative and illustrative, rather than a validated quantitative account of explanation faithfulness or clinical reliability.

At the same time, the current findings should be interpreted within the scope of the present study. The reported results are based on internal evaluation on SDD only, and the absence of external validation remains a primary limitation. Because SDD is a newly curated dataset from a limited source setting, the observed performance may partly reflect site-specific characteristics, including local imaging protocols, demographic composition, and labeling practices. In addition, patient-profile variables may be sensitive to temporal and geographic data shift, and the current dataset does not yet support detailed stage-specific analysis or comprehensive evaluation of overlapping pulmonary diseases and co-morbidities.

In practical terms, SiCLIP is best viewed as a multimodal screening- and triage-support framework for occupationally exposed populations, rather than a standalone diagnostic or deployment-ready system. Future work should focus on external validation across institutions and regions, broader and more diverse cohorts, richer annotations, evaluation under disease overlap, and prospective assessment in real screening workflows.

CRediT authorship contribution statement

Duy Le: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Tien Nguyen:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Investigation, Data curation, Conceptualization. **Huyen Nguyen:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Formal analysis. **Anh Nguyen:** Visualization, Validation, Supervision, Resources, Data curation. **Cong Tran:** Writing – review & editing, Validation, Supervision, Resources, Project administration. **Cuong Pham:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Investigation.

Ethics approval and consent to participate

This study utilizes data collected under the National Scientific Research Project titled “Study on Molecular Epidemiological Characteristics, Risk Factors, and the Application of Advanced Techniques in Early Diagnosis of Silicosis in Vietnam” (Project Code: KC.10.33/16-20). The original study received ethical approval from the Institutional Review Board (IRB) of Hanoi Medical University, under approval number 4218/HMUIRB, issued on November 16, 2018, by the Institute of Preventive and Public Health, Hanoi Medical University.

All data were collected in accordance with national regulations and international ethical guidelines, including the Declaration of Helsinki. Written informed consent was obtained from all participants prior to data collection. Participants were fully informed about the study objectives, procedures, potential risks, and their rights, including the right to withdraw at any time without consequences.

For this study, the dataset was shared with the authors with explicit permission from the principal investigators of the original research project. As all data were fully anonymized before access, no additional ethical approval was required for this secondary analysis.

Importantly, to the best of our knowledge, SDD is the **first multimodal silicosis dataset in Viet Nam** that pairs chest radiographs with structured worker-profile information, including occupational exposure history, harmful habits, and clinically relevant symptoms. This makes SDD a distinctive national resource for studying multimodal screening of occupational silicosis and silica-related lung-health risks.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.artmed.2026.103456>.

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